

R Programming – Standard Questions

Scatterplots and Correlograms

Instructions: Answer all five questions using R programming. No answers are provided.

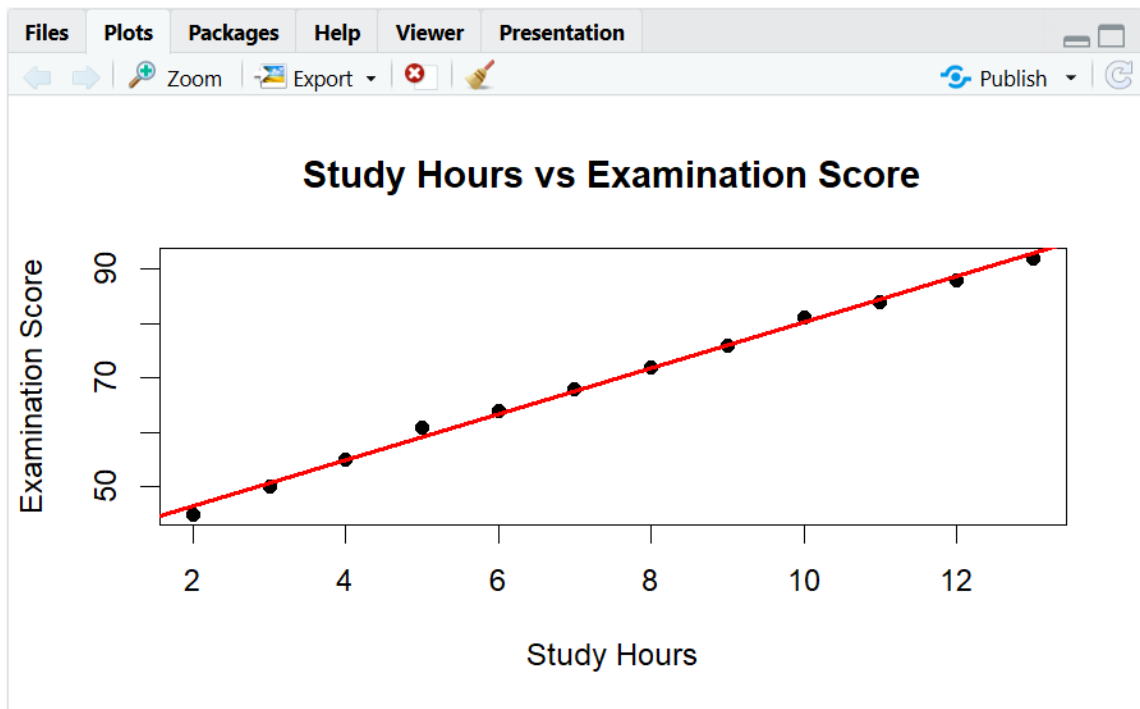
Question 1 – Scatterplot Analysis

A researcher records the following data for 12 students: Study_Hours = c(2,3,4,5,6,7,8,9,10,11,12,13) and Exam_Score = c(45,50,55,61,64,68,72,76,81,84,88,92).

Using R, construct an appropriate scatterplot to examine the relationship between study hours and examination score. Customize the plot with suitable axis labels, title, plotting symbols, and a fitted regression line.

Code:

```
Study_Hours <- c(2,3,4,5,6,7,8,9,10,11,12,13)
Exam_Score <- c(45,50,55,61,64,68,72,76,81,84,88,92)
plot(Study_Hours, Exam_Score,
     main = "Study Hours vs Examination Score",
     xlab = "Study Hours",
     ylab = "Examination Score",
     pch = 19)
model <- lm(Exam_Score ~ Study_Hours)
abline(model, col = "red", lwd = 2)
cor(Study_Hours, Exam_Score)
```



[1] 0.9982654

Question 2 – Scatterplot with Grouping

A company studies the relationship between Advertising_Expenditure and Sales for 15 observations. The observations belong to three regions: North, South, and West.

Using R, create a grouped scatterplot that represents Advertising_Expenditure on the x-axis and Sales on the y-axis, with different visual representations for the three regions. Add an appropriate title, legend, and axis labels. Briefly assess the strength and direction of the relationships visible in the plot.

Code:

```
Advertising_Expenditure <- c(
```

```
  10, 12, 15, 18, 20,
```

```
  11, 14, 16, 19, 22,
```

```
  9, 13, 17, 21, 24
```

```
)
```

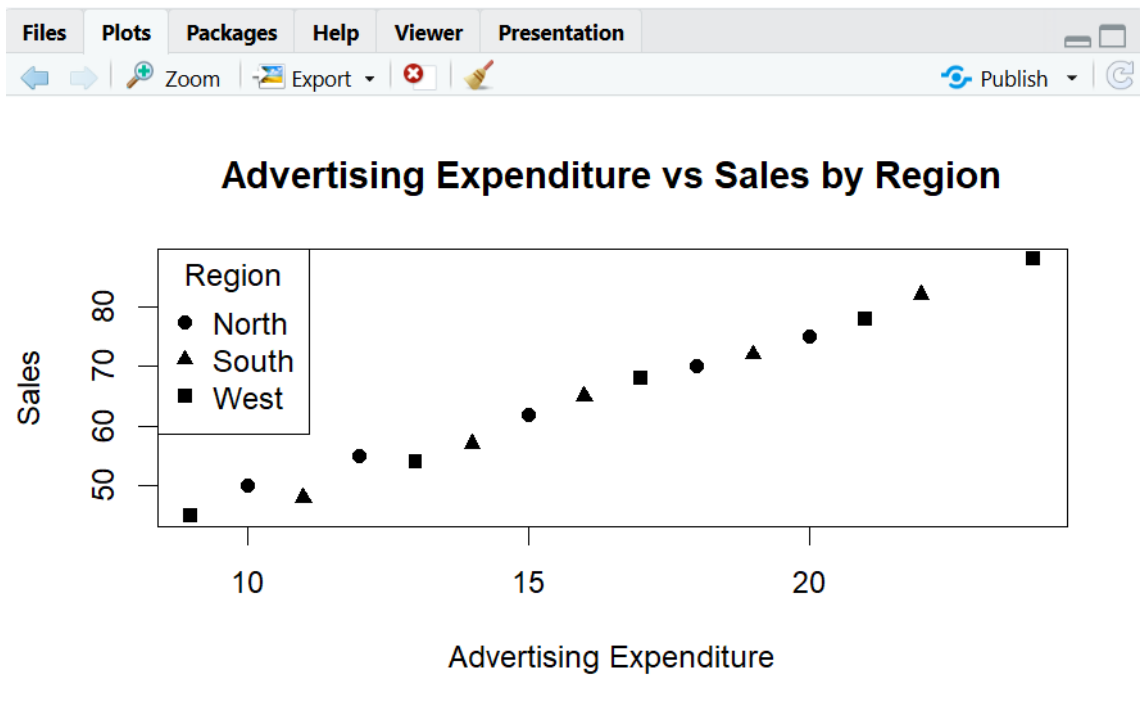
```
Sales <- c(
```

```
  50, 55, 62, 70, 75,
```

```
  48, 57, 65, 72, 82,
```

```
  45, 54, 68, 78, 88)
```

```
Region <- c(
  "North", "North", "North", "North", "North",
  "South", "South", "South", "South", "South",
  "West", "West", "West", "West", "West"
)
Region <- factor(Region)
symbols <- c(19, 17, 15)
plot(Advertising_Expenditure, Sales,
     pch = symbols[Region],
     main = "Advertising Expenditure vs Sales by Region",
     xlab = "Advertising Expenditure",
     ylab = "Sales")
legend("topleft",
     legend = levels(Region),
     pch = symbols,
     title = "Region")
```



```
cor(Advertising_Expenditure[Region == "North"],
```

```
    Sales[Region == "North"])
```

```
cor(Advertising_Expenditure[Region == "South"],
```

```
    Sales[Region == "South"])
```

```
cor(Advertising_Expenditure[Region == "West"],
```

```
    Sales[Region == "West"])
```

```
> # Correlation for each region
> cor(Advertising_Expenditure[Region == "North"],
+     Sales[Region == "North"])
[1] 0.9997648
>
> cor(Advertising_Expenditure[Region == "South"],
+     Sales[Region == "South"])
[1] 0.9979095
>
> cor(Advertising_Expenditure[Region == "West"],
+     Sales[Region == "West"])
[1] 0.9979093
> |
```

Question 3 – Correlation and Correlogram

The following dataset contains measurements of five variables for 10 observations: Temperature, Humidity, Pressure, Wind_Speed, and Rainfall.

Using R, calculate the correlation matrix and construct a correlogram to visualize the pairwise correlations among all variables. Use an appropriate visualization scale to distinguish positive and negative correlations and display the correlation values where suitable.

Code:

```
install.packages("corrplot")

library(corrplot)

data <- data.frame(

  Temperature = c(25,27,29,30,32,34,35,28,26,31),

  Humidity = c(80,76,72,68,65,60,58,75,78,66),

  Pressure = c(1012,1010,1008,1007,1005,1003,1002,1009,1011,1006),

  Wind_Speed = c(10,12,14,15,18,20,22,13,11,17),

  Rainfall = c(12,10,8,7,5,3,2,9,11,6)

)

cor_matrix <- cor(data)

print(cor_matrix)

corrplot(cor_matrix,

  method = "color",

  type = "upper",

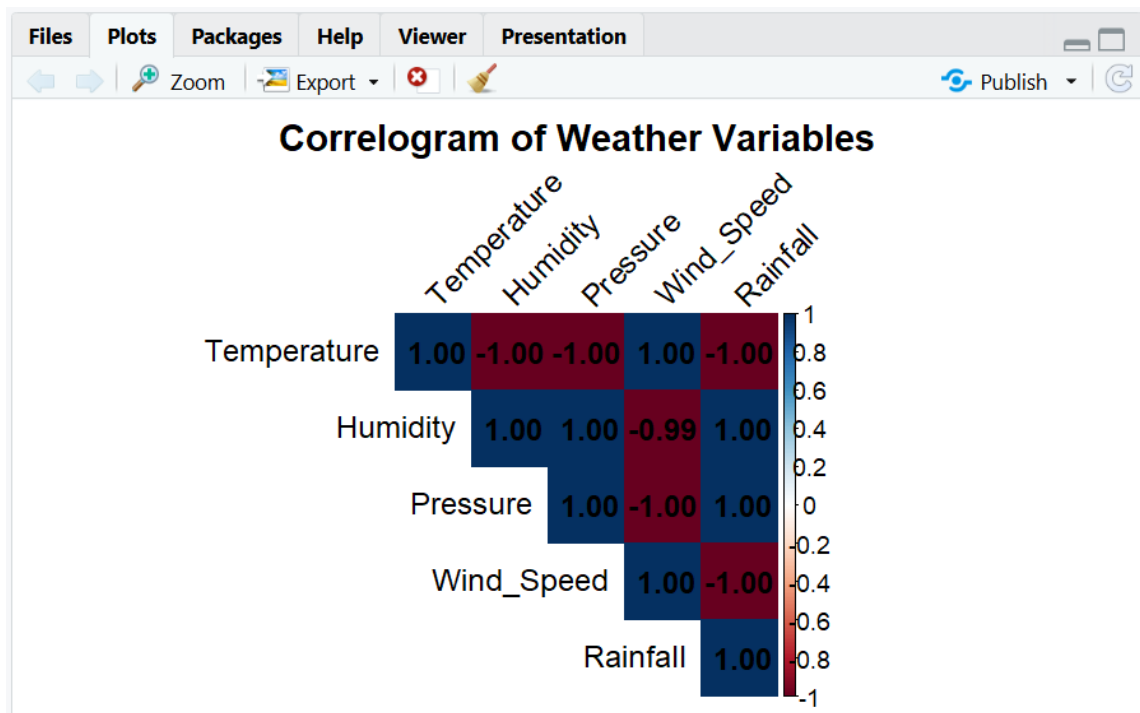
  addCoef.col = "black",

  tl.col = "black",

  tl.srt = 45,

  title = "Correlogram of Weather Variables",

  mar = c(0,0,2,0))
```



Question 4 – Correlogram for Multivariate Data

A university collects data on Attendance, Assignment_Score, Internal_Marks, Study_Hours, and Final_Marks for 12 students.

Using R, generate a correlogram showing the pairwise relationships among these variables. Apply suitable labels and visualization settings so that strong, moderate, and weak correlations can be distinguished clearly. Identify the variable that appears to have the strongest relationship with Final_Marks based on the visualization.

Code:

```
install.packages("corrplot")

library(corrplot)

data <- data.frame(

  Attendance = c(70,75,80,85,90,92,88,78,82,95,96,98),

  Assignment_Score = c(60,65,70,75,80,85,82,68,72,88,90,94),

  Internal_Marks = c(55,60,65,72,78,82,80,63,70,85,88,92),

  Study_Hours = c(2,3,4,5,6,7,6,3,4,8,9,10),

  Final_Marks = c(58,63,68,75,82,87,84,66,72,91,94,97)

)
```

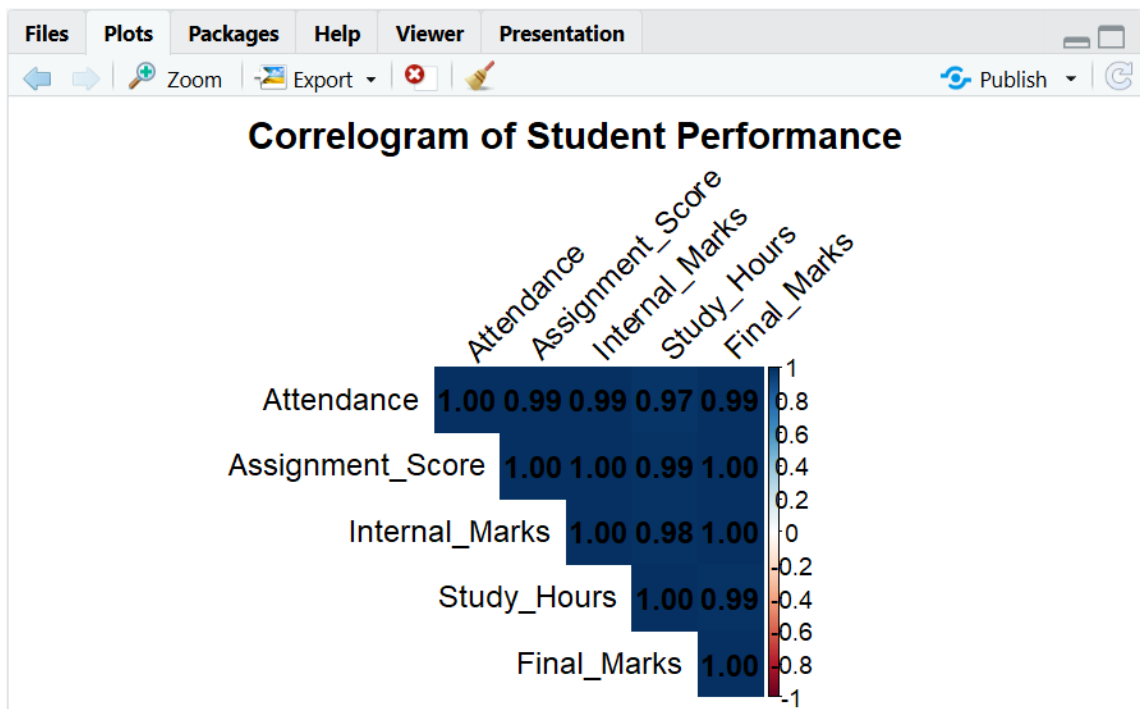
```

cor_matrix <- cor(data)

print(cor_matrix)

corrplot(cor_matrix,
         method = "color",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black",
         tl.srt = 45,
         title = "Correlogram of Student Performance",
         mar = c(0,0,2,0))

```



```

> # Correlations with Final Marks
> final_correlations <- cor_matrix[, "Final_Marks"]
>
> print(final_correlations)
      Attendance Assignment_Score Internal_Marks Study_Hours
0.9916103      0.9985053      0.9982786      0.9858079
Final_Marks
1.0000000
>
> # Remove Final_Marks itself
> final_correlations <- final_correlations[
+   names(final_correlations) != "Final_Marks"
+ ]
>
> # Find strongest relationship
> strongest <- names(which.max(abs(final_correlations)))
>
> print(strongest)
[1] "Assignment_Score"
> print(final_correlations[strongest])
Assignment_Score
0.9985053

```

Question 5 – Integrated Scatterplot and Correlogram Analysis

A retail dataset contains 15 observations with the variables Price, Advertising, Customer_Visits, Discount, and Sales.

Using R, perform the following:

- Construct suitable scatterplots to examine the relationships of Sales with Price, Advertising, Customer_Visits, and Discount.
- Compute the correlation matrix for all numerical variables.
- Create a correlogram to visualize the correlation structure.
- Compare the scatterplots and correlogram and identify any strong positive or negative relationships and possible variables that may require further investigation.

Code:

```

install.packages("corrplot")

library(corrplot)

data <- data.frame(

  Price = c(50,55,60,65,70,75,80,85,90,95,100,105,110,115,120),

  Advertising = c(

    10,12,14,16,18,

    20,22,24,26,28,

    30,32,34,36,38

  ),

```



```

Customer_Visits = c(
  120,125,130,138,145,
  150,158,165,170,178,
  185,192,200,208,215
),
Discount = c(
  5,6,7,8,10,
  11,12,14,15,16,
  18,19,21,22,25
),
Sales = c(
  100,108,115,125,132,
  140,148,155,165,172,
  180,188,198,208,220
)
)
print(data)
plot(data$Price, data$Sales,
  main = "Sales vs Price",
  xlab = "Price",
  ylab = "Sales",
  pch = 19)
abline(lm(Sales ~ Price, data = data),
  col = "red", lwd = 2)

```

```
plot(data$Advertising, data$Sales,
      main = "Sales vs Advertising",
      xlab = "Advertising",
      ylab = "Sales",
      pch = 19)
abline(lm(Sales ~ Advertising, data = data),
      col = "red", lwd = 2)
plot(data$Customer_Visits, data$Sales,
      main = "Sales vs Customer Visits",
      xlab = "Customer Visits",
      ylab = "Sales",
      pch = 19)
abline(lm(Sales ~ Customer_Visits, data = data),
      col = "red", lwd = 2)
plot(data$Discount, data$Sales,
      main = "Sales vs Discount",
      xlab = "Discount",
      ylab = "Sales",
      pch = 19)
abline(lm(Sales ~ Discount, data = data),
      col = "red", lwd = 2)
cor_matrix <- cor(data)
print(cor_matrix)
corrplot(cor_matrix,
          method = "color",
          type = "upper",
```

```

addCoef.col = "black",

tl.col = "black",

tl.srt = 45,

title = "Correlogram of Retail Variables",

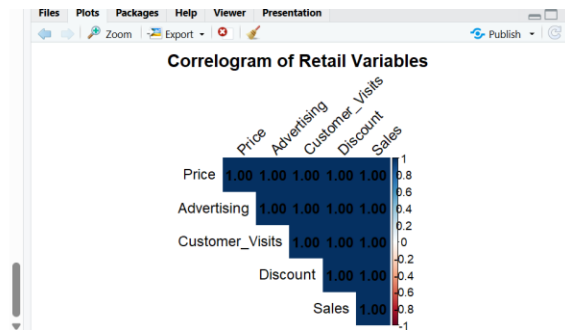
mar = c(0,0,2,0))

```

```

> print(cor_matrix)
      Price Advertising Customer_Visits Discount Sales
Price 1.0000000 1.0000000 0.9990744 0.9962410 0.9988511
Advertising 1.0000000 1.0000000 0.9990744 0.9962410 0.9988511
Customer_Visits 0.9990744 0.9990744 1.0000000 0.9978354 0.9992639
Discount 0.9962410 0.9962410 0.9978354 1.0000000 0.9981147
Sales 0.9988511 0.9988511 0.9992639 0.9981147 1.0000000
>
> # -----
> # (c) Correlogram
> # -----
>
> corplot(cor_matrix,
+         method = "color",
+         type = "upper",
+         addCoef.col = "black",
+         tl.col = "black",
+         tl.srt = 45,
+         title = "Correlogram of Retail Variables",
+         mar = c(0,0,2,0))
>

```



```

b)sales_correlations <- cor_matrix[, "Sales"]

```

```

print(sales_correlations)

```

```

> sales_correlations <- cor_matrix[, "Sales"]
>
> print(sales_correlations)
      Price Advertising Customer_Visits Discount
0.9988511 0.9988511 0.9992639 0.9981147
Sales
1.0000000
> |

```